

Page-by-page Script

What every screen of the NexLink console shows, what to point at, and what to say

Item	Detail
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Prepared on	21 October 2026
Length	About 20 minutes for all 12 pages; each page works on its own
Data on screen	Demo scenario created by scripts\demo-data.ps1 (real enrollments, a real tunnel, [TEST] drills) plus your own pc-1
Companion documents	NexLink-Demo-Guide.pdf (full demo with commands), run-commands.md (start-up)

Contents

Before you start 3
Sign in and sign up 4
Overview 5
Networks 6
Devices 7
Device detail 8
Topology 10
Simulation 11
Diagnostics 12
Incident detail 13
Approvals 14
Audit log 15
Register a PC 16
Suggested order and timing 17

Before you start

1. Start everything

Double-click scripts\start-all.cmd (or run the five commands in docs\run-commands.md). It starts the database, control plane, AI service, web console, the pc-1 agent and — if demo data exists — the demo tunnel window. Then open http://localhost:3000 and sign in as the admin account.

2. Create the demo data (once)

```
cd D:\projects\cn
.\scripts\demo-data.ps1          # create
.\scripts\demo-data.ps1 -Remove  # delete it again
```

Nothing is typed into the database by hand. The script uses the real system end to end, so every page shows genuine records:

Demo item	Shows as	Why it exists
demo-laptop-a / demo-laptop-b	healthy · connected	Enrolled by the real agent; a real WireGuard tunnel runs between them in the "demo tunnel" window
demo-desktop	unknown	Enrolled but never started — shows that NexLink says “unknown”, not healthy
demo-old-laptop	revoked	Enrolled, then revoked (“laptop reported lost”)
demo-printer	pending enrollment	Registered, but its one-time token was never used
demo-campus / demo-branch	networks	10.88.0.0/24 and 10.89.0.0/24
3 [TEST] incidents	diagnostics	Drills through the real path: one approved test verified successful, one action awaiting approval, one rejected by policy, plus a Groq AI diagnosis

> **Be careful**
>
> **Keep the “demo tunnel” window open.** If it is closed, demo-laptop-a and -b stop reporting and become unreachable after about a minute (which is the correct behaviour — but not what you want on screen).
>
> **Be open about what is a drill.** Incidents marked [TEST] are labelled drills and the simulation page is labelled simulated. Everything else — enrollment, tunnel, measurements, approvals, audit — is real.

Sign in and sign up

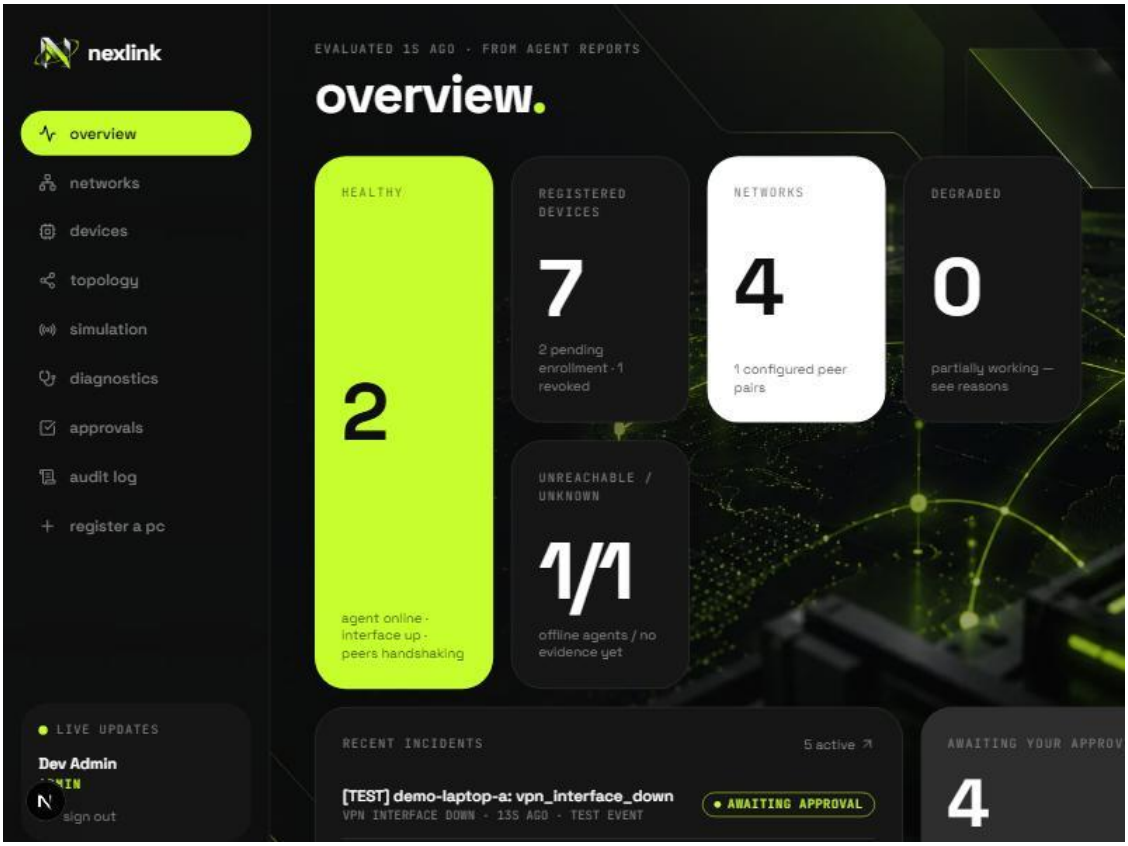
What this page is for. Every user has an account; the first account becomes the administrator, later ones are operators who see only their own networks.

- **sign in / sign up tabs** — sign up creates the account and takes you straight to “register this pc”.
- **the lime panel** — the project’s promise: nothing is marked connected until a handshake says so.

> **What to say**
> “Passwords are hashed with scrypt; the session lives in an HttpOnly, SameSite=Strict cookie, and every change needs a CSRF token. Repeated wrong passwords lock the account for 15 minutes.”

If asked	Answer
What is the difference between admin and operator?	Admin can do everything, including approving any action and managing users. Operators manage their own networks and can approve low-risk actions only. Viewers are read-only. The backend enforces this on every request.
Can someone else see my devices?	No. Each network has an owner and collaborators; anything outside your scope returns “not found”.

Overview



Overview — counts computed from agent reports, not from records.

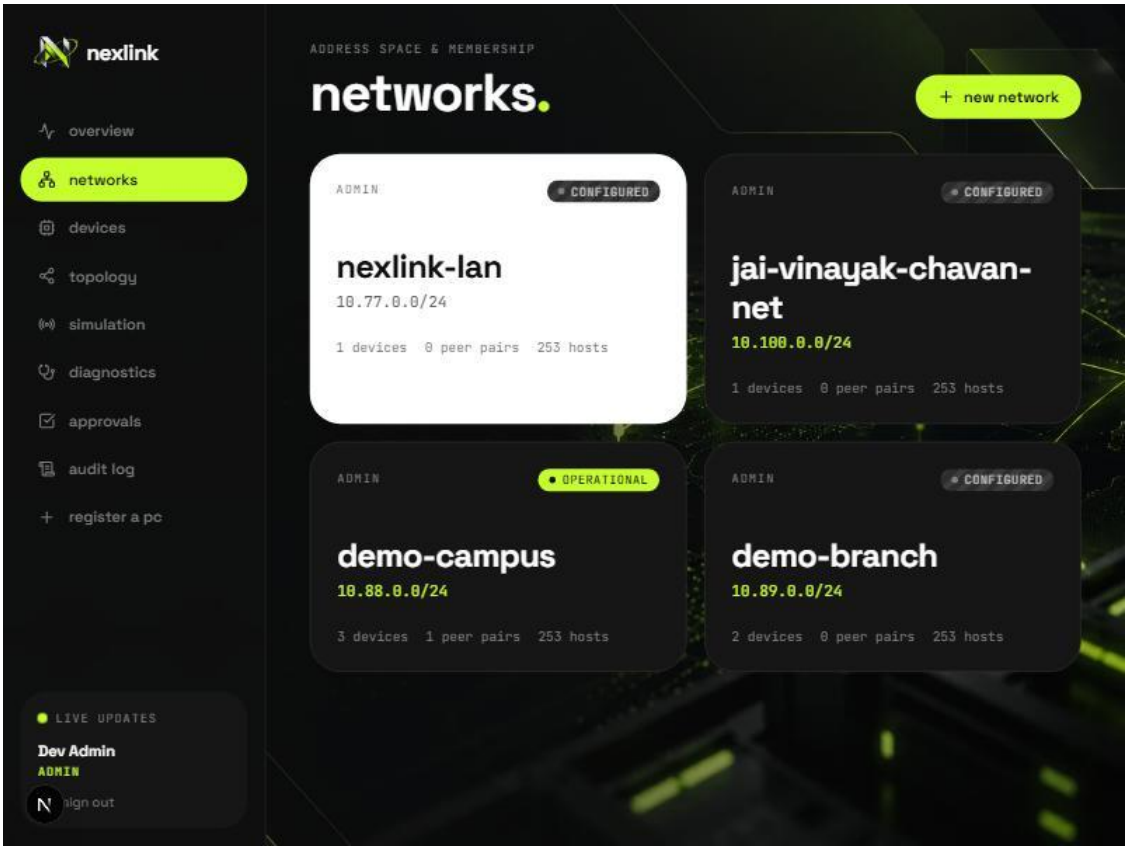
What this page is for. The one-glance health of everything you can see.

- **HEALTHY** — devices where all three signals are good: agent online, interface up, peers handshaking (the two demo laptops).
- **REGISTERED DEVICES** — every device, with pending and revoked counted separately.
- **DEGRADED** — partly working, e.g. handshake fine but high loss.
- **UNREACHABLE / UNKNOWN** — offline agents vs devices with no evidence yet (demo-desktop).
- **RECENT INCIDENTS and AWAITING YOUR APPROVAL** — shortcuts into diagnostics and approvals.
- **"evaluated … ago · from agent reports"** — the time the numbers were computed.

> **What to say**
>
> "These numbers come from what the agents measured, not from what is configured. A device that has never reported counts as unknown — never as healthy."
>
> "Health is three separate signals; the next pages show each one."

If asked	Answer
Why is a registered device not healthy?	Registered only means it has an identity. Healthy needs a recent heartbeat, the WireGuard interface up, and a fresh handshake with every peer.
How fresh is this?	Agents send a heartbeat every 30 s and telemetry every minute; the page refreshes live over Socket.IO and every 15 s.

Networks



Networks — each card is one private address space.

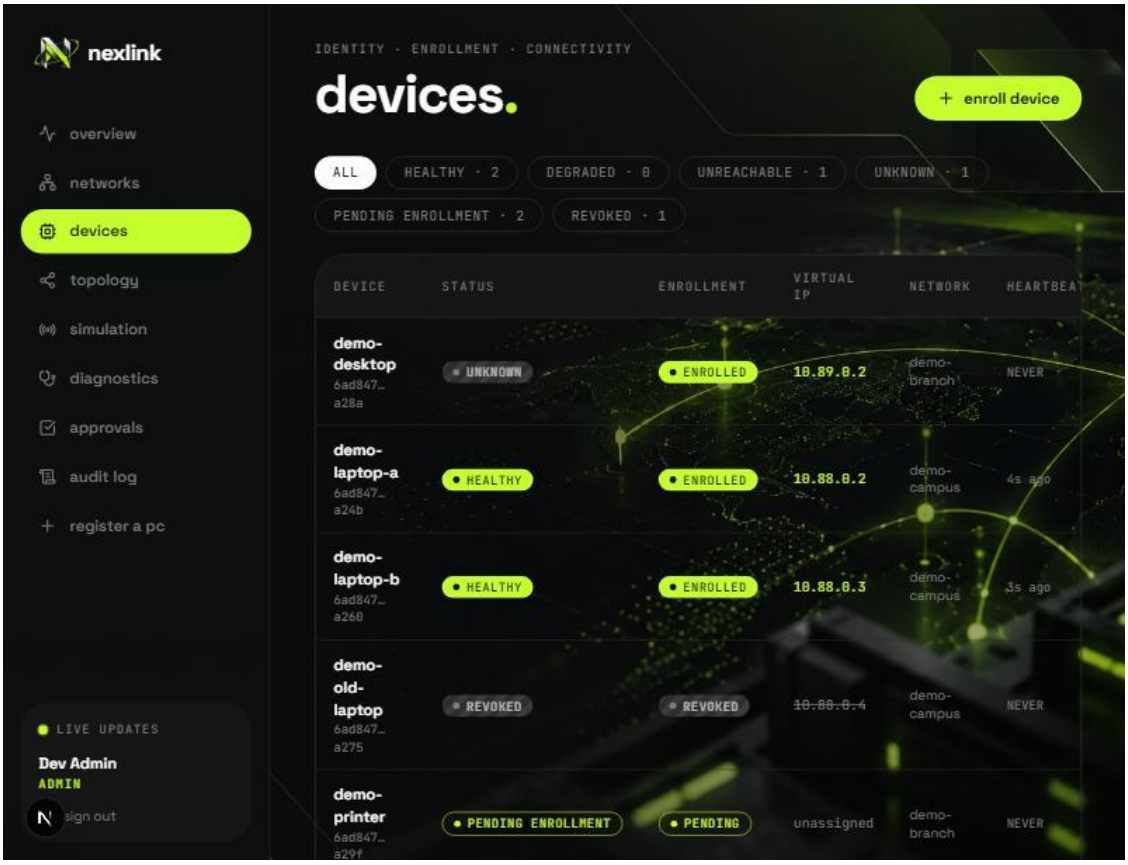
What this page is for. Private networks (address spaces) and who belongs to them.

- **CIDR (e.g. 10.88.0.0/24)** — the private range; each device gets one address from it.
- **state badge** — operational when peers are verified; configured when nothing has been observed yet.
- **devices · peer pairs · hosts** — 253 usable addresses in a /24.
- **new network** — try 10.88.0.0/25 (overlap), 10.200.0.5/24 (host bits set) and 8.8.8.0/24 (not private) to show validation.

> **What to say**
> "Address ranges are validated and checked for overlap; a unique index in MongoDB guarantees no two devices ever get the same address, even if they enroll at the same moment."

If asked	Answer
Why private ranges only?	Using public addresses inside a VPN would hijack real internet destinations for that PC.
What is a peer pair?	Two devices in the same network that should talk to each other. NexLink computes these and gives each agent its peers.

Devices



Devices — one row per PC, with the state filters on top.

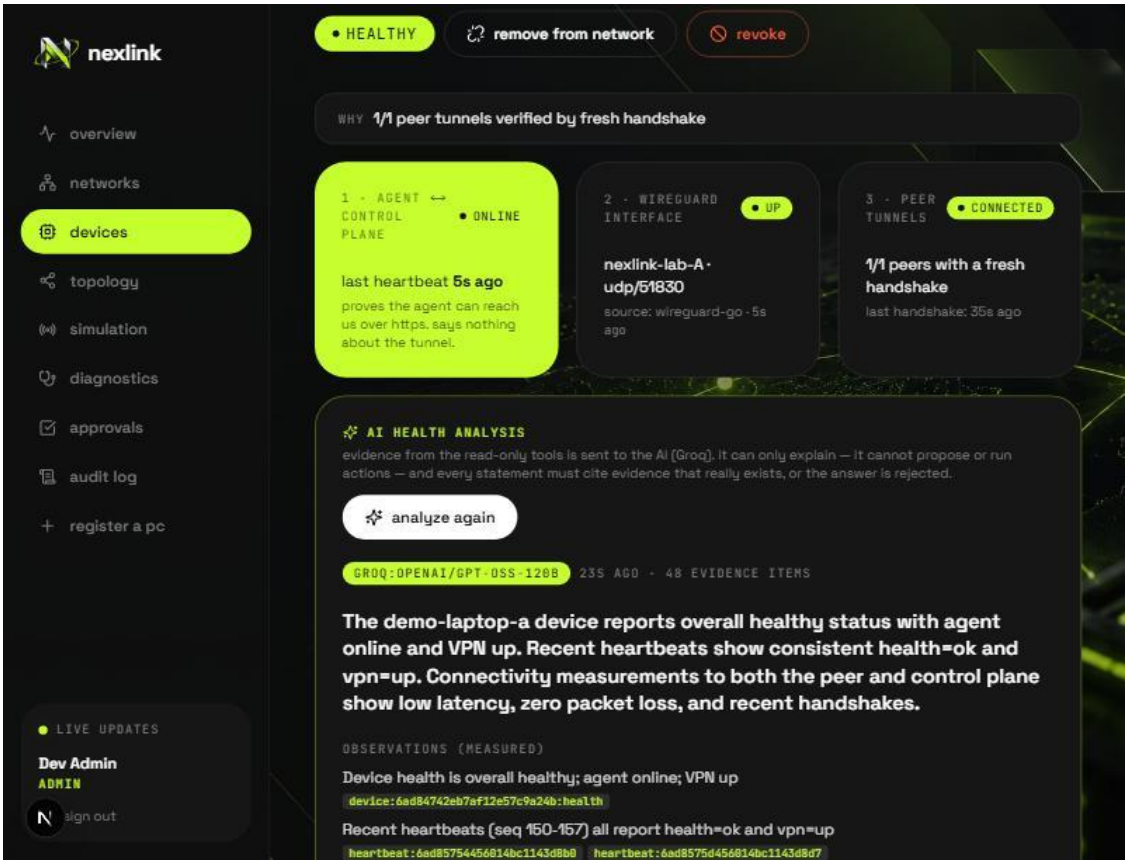
What this page is for. Every registered PC, its health and enrollment state.

- **filter chips** — healthy, degraded, unreachable, unknown, pending enrollment, revoked — each demo device shows one of them.
- **STATUS vs ENROLLMENT** — enrolled says who it is; status says how it is doing. They are deliberately separate columns.
- **VIRTUAL IP** — the address inside the private network (struck through when revoked).
- **HEARTBEAT** — “never” for demo-desktop: it enrolled but its agent was never started.
- **enroll device** — issues a one-time token for a new PC.

> **What to say**
>
> “Every PC generated its own keys during enrollment. The server only stores public keys and a fingerprint — private keys never leave the device.”

If asked	Answer
What does pending enrollment mean?	A token was issued but no agent has used it yet. Tokens are single-use, expire after 15 minutes and are stored only as hashes.
What happens to a revoked device?	Its next signed request is rejected (403 DEVICE_REVOKED) and the other devices remove it from their WireGuard peers.

Device detail



demo-laptop-a — the three independent signals, identity and observed peer links.

What this page is for. Everything about one PC: why it has its status, its identity, its tunnels, and a live view of its agent.

- **WHY line** — the reason for the status in plain words (“1/1 peer tunnels verified by fresh handshake”).
- **1 · agent ↔ control plane** — last heartbeat; proves the agent reaches the server, says nothing about the tunnel.
- **2 · WireGuard interface** — interface name, UDP port and source (wireguard-go for the userspace agent).
- **3 · peer tunnels** — how many peers have a fresh handshake.
- **AI health analysis → analyze with AI** — Groq reads this PC’s evidence and explains its health in 5–15 s: summary, observations citing evidence ids, hypotheses, uncertainties. It can only explain; it is given no actions to propose.
- **identity & membership** — SHA-256 fingerprints of the identity and WireGuard keys, virtual IP, agent version.
- **peer links (observed)** — each peer, the path (direct) and handshake age.
- **agent console (scroll down)** — the agent’s own log lines, streamed live: “telemetry sent”, “tunnel probe … rtt_ms=…”.
- **run on this device** — connectivity test, test tunnel to a peer, refresh status; each request goes to approvals.

What to say

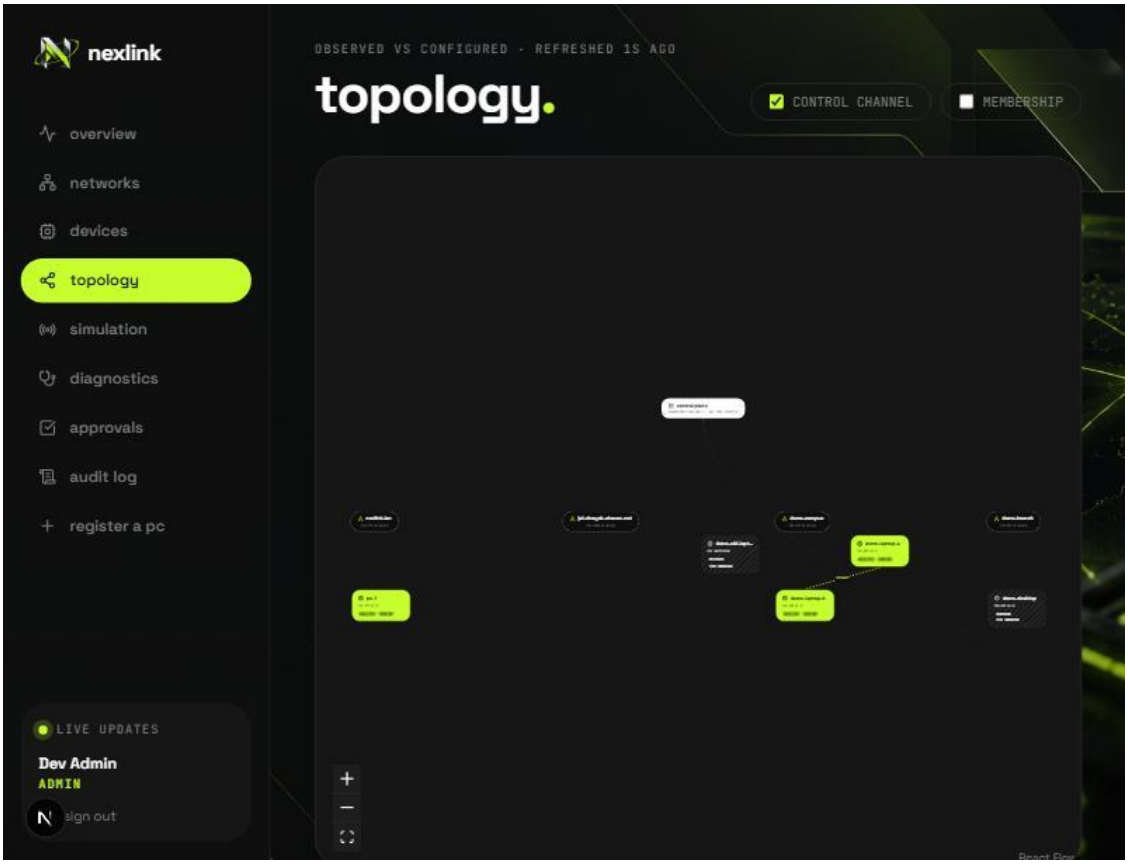
“Three cards, three independent facts. A heartbeat does not prove the tunnel works — only a WireGuard handshake does.”

“The console at the bottom is the agent on that PC talking to us; no terminal needed.”

“Analyze with AI sends only evidence summaries to Groq. Every sentence it writes must cite real evidence — if it invents a reference or tries to propose an action, the answer is rejected.”

If asked	Answer
Is the RTT real?	Yes. The agent opens TCP connections to the peer's tunnel address through the encrypted tunnel and measures them.
Can I change the PC's network settings from here?	Only if privileged mode was switched on locally on that PC. The server cannot enable it.

Topology



Topology — control plane, networks and devices; edge style shows what has been observed.

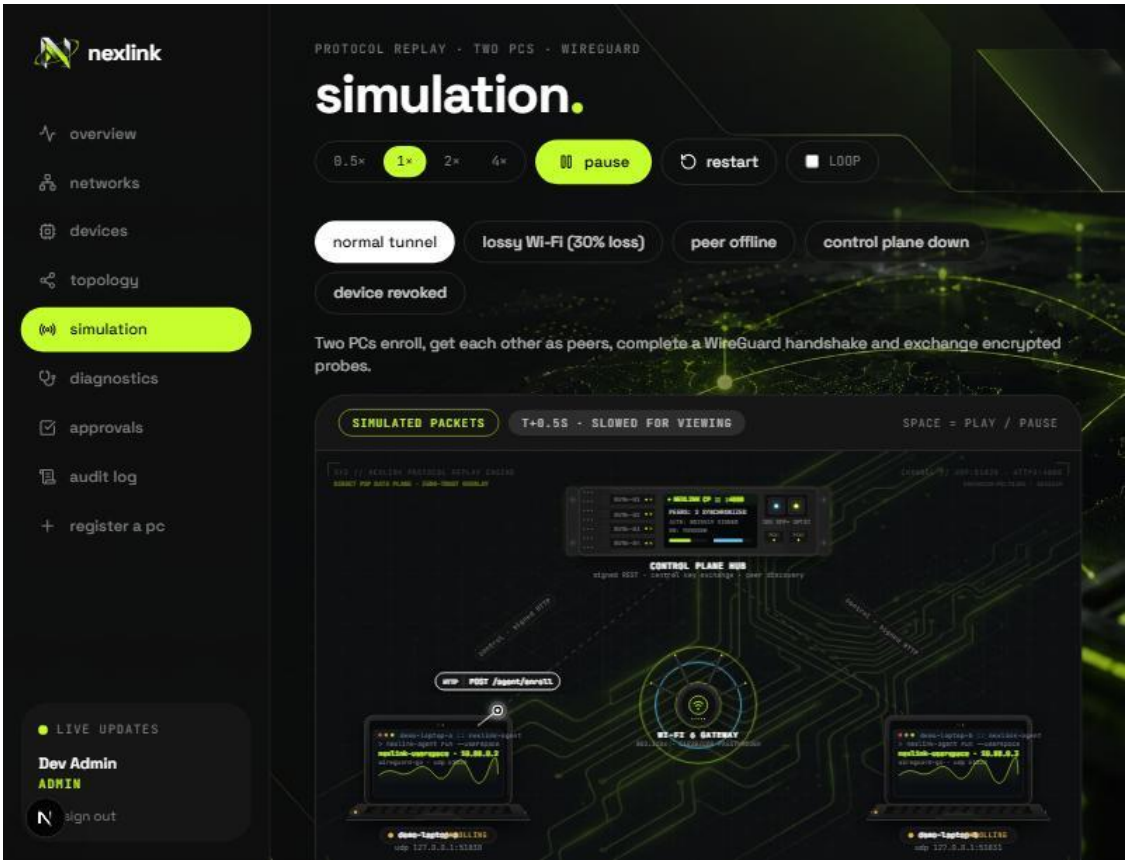
What this page is for. A live map of networks, devices and links, drawn from evidence.

- **solid lime edge** — connected: a fresh WireGuard handshake was reported (demo-laptop-a ⇄ demo-laptop-b).
- **dashed lime / dashed coral** — connecting / unreachable (handshake went stale).
- **dotted grey** — configured only: the peers should talk but nothing has been observed.
- **control channel / membership toggles** — show the agent ↔ server links and which network each device belongs to.
- **verified tunnels n/m** — counter on the right; zoom with the mouse wheel or the + / − buttons.

> **What to say**
> "A link turns lime only because WireGuard reported a handshake — not because a record exists. Configured but unproven links stay grey."

If asked	Answer
What does "direct" mean?	The tunnel goes straight between the two PCs over UDP. Traffic never passes through the server; a relay path is designed but not used.

Simulation



Simulation — the exact protocol sequence, slowed down (labelled SIMULATED PACKETS).

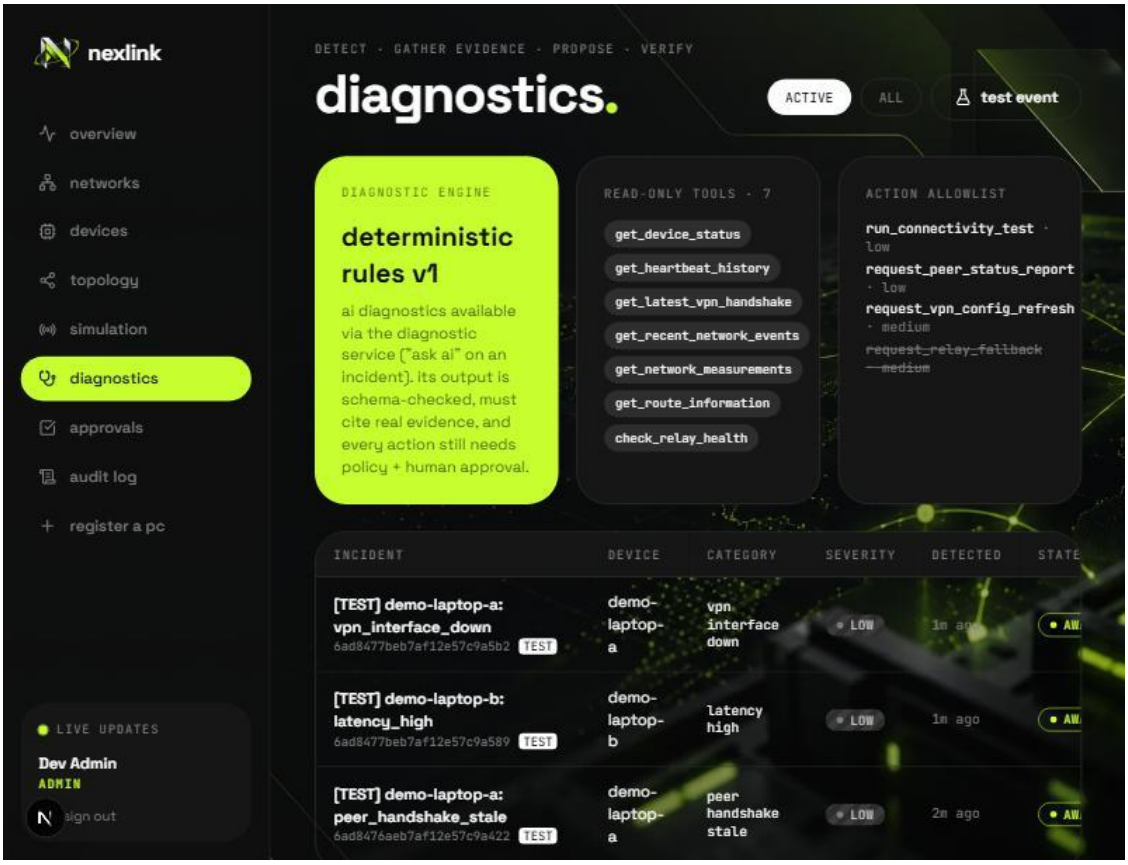
What this page is for. Explains how the tunnel is built, step by step, before you show the real one.

- **scenario buttons** — normal tunnel, lossy Wi-Fi, peer offline, control plane down, device revoked.
- **phase chips** — enrollment → peer config → handshake → encrypted traffic → telemetry → scenario event; click to jump.
- **packet capsules** — HTTP (signed control traffic) and WG-INIT 148 B, WG-RESP 92 B, WG-DATA, KEEPALIVE 32 B.
- **packet inspector / event log** — real header fields and agent-style log lines.
- **real link panel** — when two enrolled devices are selected, what their agents actually report.
- **space bar** — play / pause; speed 0.5×–4×.

> **What to say**
>
> "This is a replay of our protocol — every packet is labelled simulated — but the messages, sizes and endpoints are the real ones."
>
> "Watch control plane down: the server goes red, telemetry fails, yet the tunnel keeps working. That is why control plane and data plane are separate."

If asked	Answer
Is this real traffic?	No, it is a replay. The real tunnel is on topology and the device pages; this page explains it.

Diagnostics



Diagnostics — engine, read-only tools, the action allowlist and the incidents.

What this page is for. Where problems are detected and investigated.

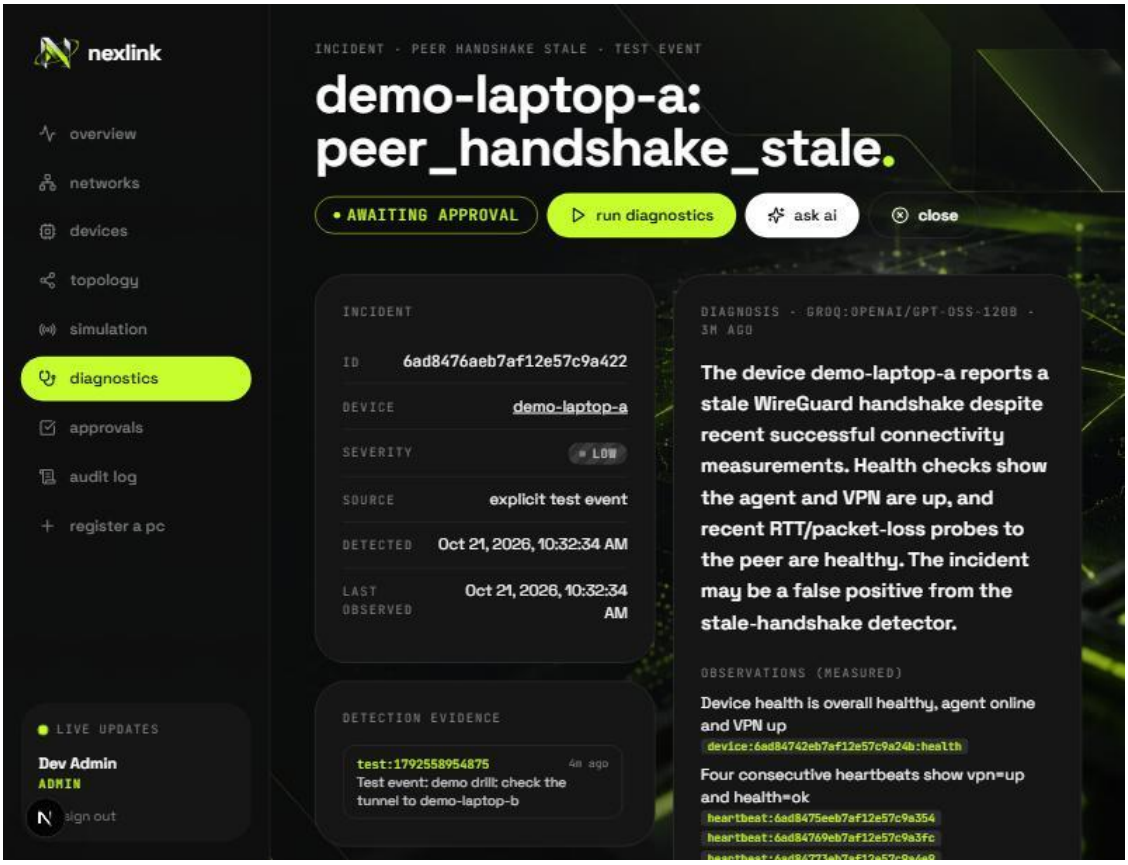
- **diagnostic engine** — deterministic rules; AI diagnosis is available via the diagnostic service (“ask ai” on an incident).
- **read-only tools (7)** — the only things a diagnosis may use: device status, heartbeat history, last handshake, events, measurements, routes, relay health.
- **action allowlist** — everything that can ever be run, with its risk; struck-through actions are not available.
- **incidents table** — [TEST] marks drills; real incidents (e.g. “pc-1: agent heartbeats stopped”) are raised automatically every 30 s.
- **test event** — create a labelled drill without breaking a network.

What to say

“Incidents are detected from measurements. Diagnosis can only read; anything that changes something must come from this allowlist and pass the policy engine and a human.”

If asked	Answer
Do incidents close themselves?	Real ones resolve automatically when fresh measurements show the problem is gone. Drills never auto-resolve.

Incident detail



An incident with a Groq AI diagnosis — every observation cites evidence ids.

What this page is for. One incident: evidence, diagnosis, proposed actions and their outcome.

- **run diagnostics / ask ai** — deterministic rules, or the AI provider (the label shows which, e.g. groq:openai/gpt-oss-120b).
- **observations (measured)** — each line cites evidence ids such as heartbeat:… or device:…:health.
- **hypotheses / uncertainties** — marked as untested; the AI must say what it does not know.
- **proposed actions** — each with the policy verdict; approve & run executes on the agent and shows the verified outcome.
- **evidence (right)** — everything the read-only tools collected; cited items are highlighted.

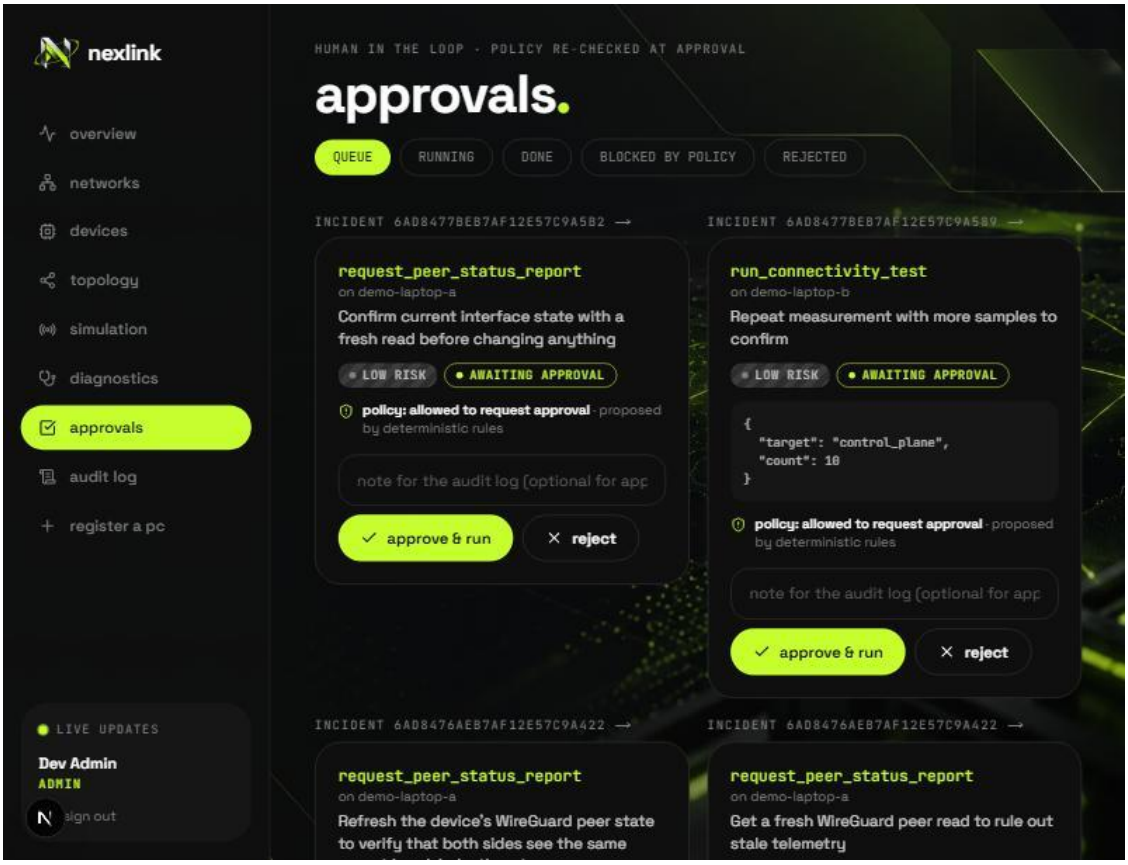
What to say

"The AI only receives evidence summaries, never IP addresses. If it cites evidence that does not exist, gives a confidence score or proposes an action outside the allowlist, the answer is rejected."

"The approved connectivity test here was executed by the agent through the tunnel and verified from fresh measurements."

If asked	Answer
What if the AI is wrong or offline?	Wrong output fails validation and is discarded; deterministic rules always work; the VPN never depends on the AI.

Approvals



Approvals — the human-in-the-loop queue.

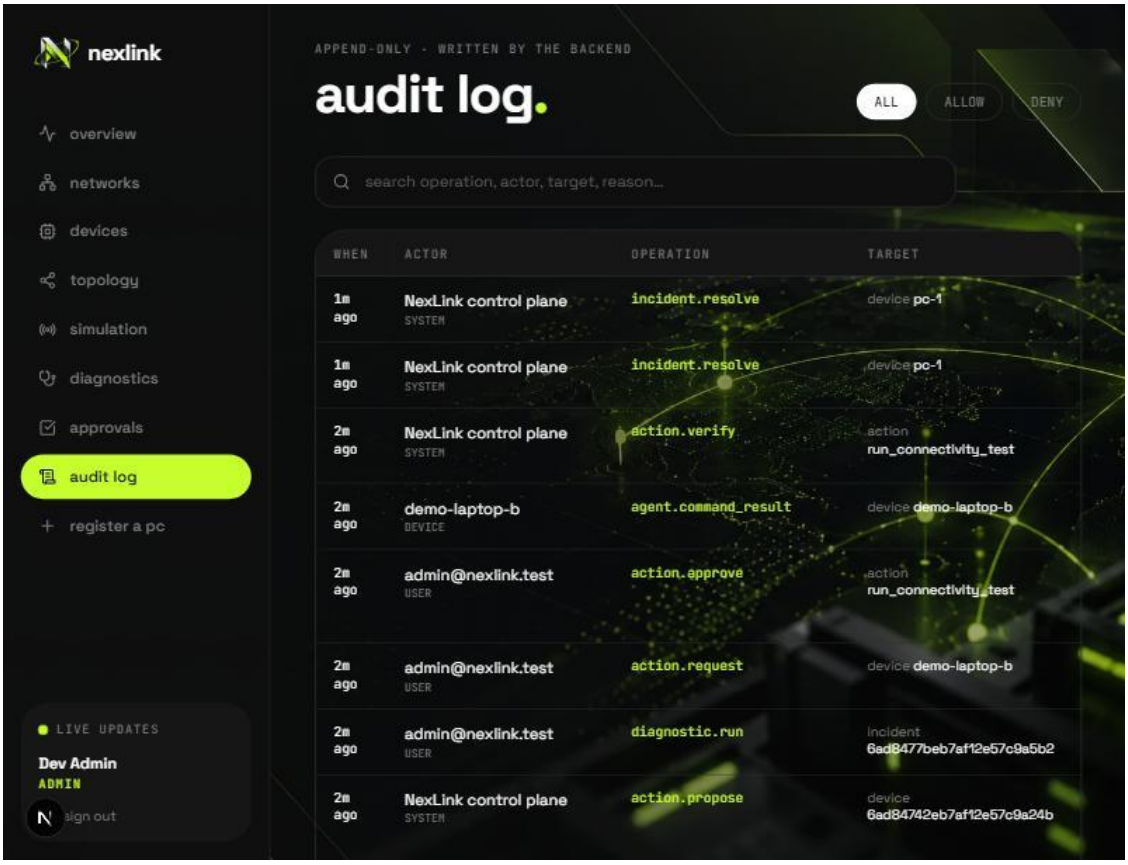
What this page is for. Every action that would run on a PC waits here for a human decision.

- **tabs** — queue, running, done, blocked by policy, rejected.
- **action card** — action type, device, rationale, risk, the policy verdict and the exact parameters.
- **approve & run / reject** — with an optional note that goes into the audit log; policy is checked again at approval time.
- **blocked by policy** — e.g. request_vpn_config_refresh on demo-laptop-a: privileged mode is off on that PC.

> **What to say**
> "Nothing runs on a PC without passing the policy engine and a human approval — and the agent checks its own allowlist again before executing."

If asked	Answer
Can an operator approve everything?	No. Operators can approve low-risk actions only; medium risk needs an admin.

Audit log



Audit log — written by the backend; there is no API to edit or delete it.

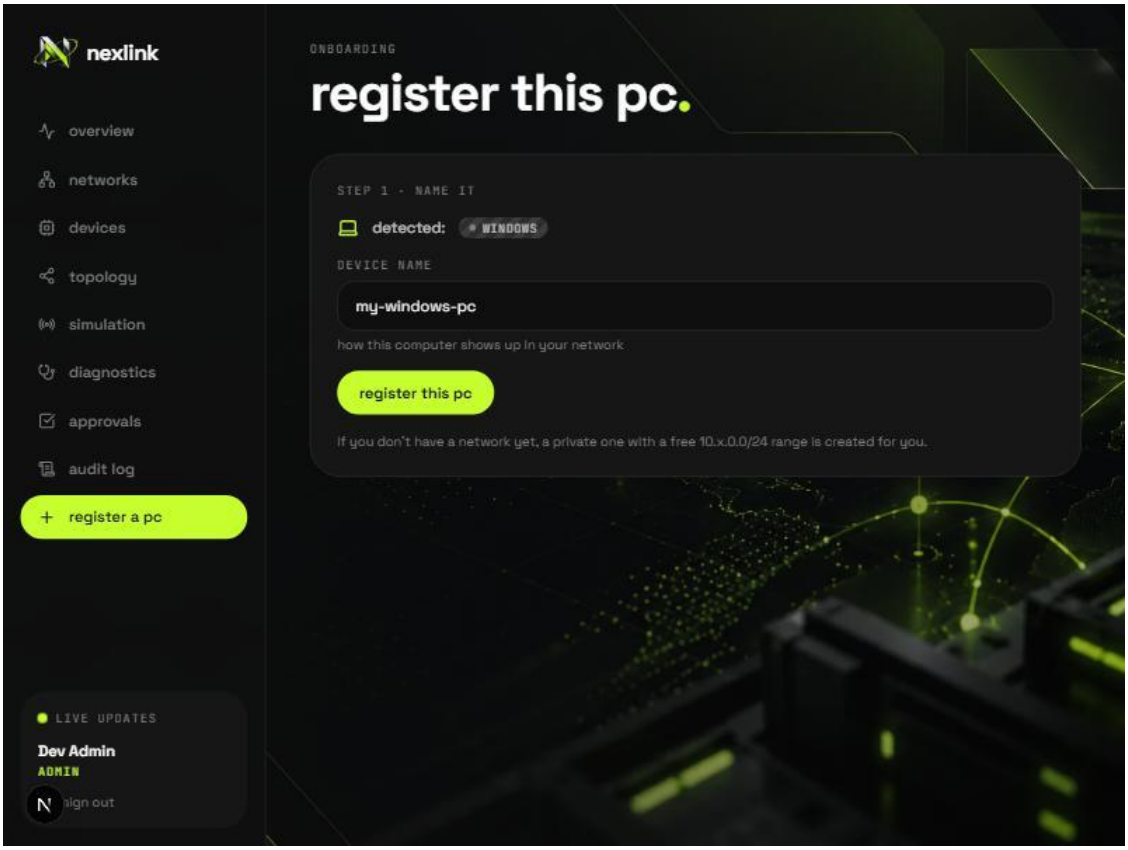
What this page is for. A tamper-evident history of every security-relevant event.

- **actor** — a USER (admin@…), a DEVICE (an agent) or the SYSTEM (control plane, e.g. action.verify, incident.resolve).
- **operation** — device.enroll, device.revoke, diagnostic.run, action.request / approve / verify, agent.command_result …
- **allow / deny filter and search** — denied entries show refused requests, e.g. a revoked device trying to sign in.

> **What to say**
> "The backend writes every sensitive operation here. Secrets are redacted, and the log is append-only from the API's point of view."

If asked	Answer
Why keep entries for deleted demo data?	An audit log must not be editable; removing the demo data deliberately leaves its history.

Register a PC



Register this PC — step 1; step 2 shows a single command to paste.

What this page is for. Adding a computer to a network in about a minute.

- **step 1 · name it** — the platform is detected; a private network is created if you have none.
- **step 2 · connect this pc** — copy one PowerShell line and paste it on the PC: it downloads the agent, creates the keys locally, enrolls and starts the userspace tunnel.
- **step 3 · waiting** — the page flips to "… is in" by itself when the agent proves it holds its key.

> **What to say**
>
> "The PC needs no VPN app and no administrator rights: the agent carries the WireGuard tunnel itself. The token is single-use and only travels in that PowerShell window."

If asked	Answer
What if the second PC is on another network?	It must reach this PC on the LAN (same Wi-Fi or hotspot). Across different home routers a relay is needed — designed, not part of this demo.

Suggested order and timing

#	Page	Time	One-line message
1	Sign in	1 min	Accounts, roles, secure sessions
2	Overview	2 min	Numbers come from measurements
3	Networks	1 min	Validated private address spaces
4	Devices	2 min	Every state, separately tracked
5	Device detail	3 min	Three signals + live agent console
6	Topology	1 min	Lime only after a real handshake
7	Simulation	3 min	How the tunnel is built; data plane survives a server outage
8	Diagnostics	1 min	Detect from evidence, read-only tools
9	Incident detail	3 min	AI with guardrails, verified outcome
10	Approvals	1 min	Human in the loop
11	Audit log	1 min	Everything is recorded
12	Register a PC	1 min	One pasted line adds a computer

Live extras if time allows: approve the waiting action on the approvals page and watch it verify; on devices → demo-laptop-a click “test tunnel”; connect a second physical PC (Demo Guide, Act 5b).